
200 Hz Physics (8 ticks / decision)
PyBullet Corridor Environment
CF2X Quadcopter + PID
FPV Camera (110° FOV, −10° tilt)
160 × 160 uint8 grayscale
4 frames / step
(B, 4, 160, 160)
100 Hz Sensory
Retina & Optic Lobe (Frozen flyvis)
BoxEye (721 hexals)
Model flow/0000/000 (Δt = 10 ms)
45,669 nodes, 65 cell types, no_grad
Gradient Boundary / BC Cache
T4a-d, T5a-d
(B, 8, 721)
100 Hz Bridge
LPTC Motion Pool & Input Projection
x_{d,p} = max(0, a_{d,p}) − ā_d
s_t = Pool(x_t) ∈ ℝ^{B×8}
u_t = Proj_{I_ER}(s_t) ∈ ℝ^{B×457}
(u_{t,i} = 0 ∀i ∉ I_ER)
u_t ∈ ℝ^{B×457}
(I_ER only)
100 Hz Recurrent Core (fp32, N = 457, 17,041 edges)
FlyWire Central Complex Population
r_t = φ(v_t) ⊙ m_keep, α = σ(a)
v_{t+1} = (1 − α) ⊙ v_t + α ⊙ (r_t(W ⊙ M_FlyWire) + u_t + b)
ER (278)
Ring input
EPG (51)
Compass wedge
PEN/PEG (62)
Turn shift
Δ7 (42)
Bridge shaping
PFL3 (24)
Steering output
r_{t,I_EPG} ∈ ℝ^{51}
Heading Decoder (100 Hz)
Reads r_{t,I_EPG} only → (sin Δψ_t, cos Δψ_t)
Anatomical EB vector or learned linear
r_{4k+3,I_PFL3} ∈ ℝ^{24}
Actor Readout (25 Hz, t = 4k + 3)
Reads r_{4k+3,I_PFL3} only
μ_k = tanh(W_act r_PFL3 + b_act) ∈ [−1, 1]^4
r_{4k+3}.detach()
Value Critic (PPO, 25 Hz)
Reads r_{4k+3}.detach() ∈ ℝ^{B×457}
64 × 64 tanh MLP → V_k ∈ ℝ
Closed-Loop Command a_k = [v_fwd, v_lat, v_vert, ψ̇] held for 4 core steps / 8 physics ticks (25 Hz → 200 Hz PID)